

EXAMPLE OUTPUTS, PROFILES, AND DERIVED METRICS

In various embodiments, measurements acquired by the disclosed device are converted into one or more stored outputs that form part of a persistent user profile. Outputs may be expressed as direct measured values, normalized values, derived indices, curves, or categorical bands. The specific output format may be selected to preserve privacy while maintaining quantitative meaning and repeatability across sessions.

Distance-related outputs may include a measured distance-to-terminus value, a normalized distance index, a confidence score associated with ranging stability, an averaged distance computed from multiple samples, or a repeatability metric indicating session-to-session consistency. In certain embodiments, distance values may be mapped into standardized bins to support comparability across users or across sensing modalities.

Compliance-related outputs may include pressure-versus-time curves, expansion-versus-time curves, pressure-versus-expansion curves, stiffness indices derived from slope estimation, hysteresis measures derived from inflation/deflation response, stabilization time constants, peak pressure values at defined expansion states, and symmetry or uniformity measures derived from distributed sensing elements.

In certain embodiments, the computational system generates composite metrics combining distance and compliance measurements. Such composite metrics may represent fitment, alignment, comfort tolerance, or interaction stability parameters derived from multi-modal sensor fusion and standardized calibration routines.

Outputs may further include session quality indicators such as motion artifact flags, guidance completion scores, plausibility validation results, and sensor health diagnostics. These indicators may be used to accept, reject, or prompt repetition of measurement sessions and to improve the reliability of stored profile values.

The user profile may store baseline values, longitudinal trends, and deltas over time for both distance and compliance parameters. Trend processing may support detection of changes across repeated sessions and may be used to tailor guidance routines, adjust sensing parameters, or improve measurement robustness.

In integrated system embodiments, the derived outputs may be correlated with measurements from additional sensing nodes to generate higher-order indices, calibration mappings, or compatibility parameters. Such integration may be performed locally within an application, within a remote computational platform, or through hybrid architectures distributing computation across multiple endpoints.

In various embodiments, the computational system produces abstracted visualizations representing the derived metrics without explicit anatomical depiction. Such outputs may be provided as numerical scores, bars, curves, multi-axis charts, or symbolic representations intended to preserve privacy while enabling practical interpretation.

Accordingly, the disclosed device and associated computational system provide a flexible output framework capable of supporting simplified baseline profiling, advanced analytic configurations, and integrated multi-node profile generation while maintaining secure storage, controlled sharing, and privacy-preserving presentation of sensitive physiologic information.